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Combinatorial aspects of geometric graphs

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Abstract

As a special case of our main result, we show that for all $L > 0$, each k -nearest neighbor graph in d dimensions excludes K_h as a depth L minor if $h = \Omega(L^d)$. More generally, we prove that the overlap graphs defined by Miller, Teng, Thurston and Vavasis (1993) have this combinatorial property. By a construction of Plotkin, Rao and Smith (1994), our result implies that overlap graphs have “good” cut-covers, answering an open question of Kaklamani, Krizanc and Rao (1993). Consequently, overlap graphs can be emulated on hypercube graphs with a constant factor of slow-down and on butterfly graphs with a factor of $O(\log^* n)$ slow-down. Therefore, computations on overlap graphs, such as finite element and finite difference methods on “well-conditioned” meshes and image processing on k -nearest neighbor graphs, can be performed on hypercubic parallel machines with a linear speed-up. Our result, in conjunction with a result of Plotkin, Rao and Smith, also yields a combinatorial proof that overlap graphs have separators of sublinear size. We also show that with high probability, the Delaunay diagram, the relative neighborhood graph, and the k -nearest neighbor graph of a random point set exclude K_h as a depth L minor if $h = \Omega(L^{d/2} \log n)$. © 1998 Elsevier Science B.V.

1. Introduction

Many applications, such as finite element and finite difference methods, image processing, and pattern recognition, use graphs that are defined geometrically [5,7,17,18,24,27]. These graphs usually satisfy certain geometric conditions [3,21,22,25,29]. Thus, we can study some combinatorial properties of these graphs via our knowledge in computational geometry in order to develop efficient algorithms.

Motivated by this observation, Miller et al. [17,18] proposed a class of geometric graphs, called *overlap graphs*. The class of overlap graphs has a simple geometrical definition (see Section 2). It contains all “well-conditioned” finite element and finite difference meshes in both two and three dimensions [17,18]. It also contains k -nearest neighbor graphs and planar graphs [19,27]. Using conformal mapping and analytical geometric arguments, Miller et al. [18] showed that a d -dimensional overlap

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graph has a separator of size $O(n^{1-1/d})$. Their result generalizes the planar separator theorem of Lipton and Tarjan [15] to geometric domains and has been used in parallel scientific computing [10,11] and parallel computational geometry [8,9,27]. Recently, Spielman and Teng [26] showed that the application of this geometric separator algorithm to planar graphs finds a separator of size $1.84\sqrt{n}$, reducing the constant term in the Lipton–Tarjan’s separator theorem. It is an interesting problem to find a combinatorial characterization of overlap graphs and a combinatorial proof to their small separator theorem.

In this paper, we show that overlap graphs exclude “shallow” minors. See Section 2 for the definition of the depth of a minor as introduced by Plotkin et al. [23]. In particular, we show that for all L , each overlap graph in d dimensions excludes K_h as a depth L minor for $h = \Omega(L^d)$. In conjunction with the construction in [23], this yields the first combinatorial proof of a weaker version of the small separator theorem for overlap graphs. An important consequence of our shallow minor result is that overlap graphs have “good” cut-covers, solving an open question of Kaklamani et al. [12]. Consequently, this implies that overlap graphs can be emulated by hypercube graphs at a cost of a constant factor slow-down and by butterfly graphs with a factor of $O(\log^* n)$ slow-down. Therefore, computations based on overlap graphs, such as finite element and finite difference methods and image processing on k -nearest neighbor graphs, can be performed on hypercubic parallel machine with a linear speed-up.

Our work is motivated by the result of Plotkin et al. [23] who introduced the concept of shallow minor and also showed that a special subclass of overlap graphs, the simplicial graphs with bounded aspect ratio in d dimensions (Miller and Thurston [20]), exclude K_h as a depth L minor when $h = \Omega(L^d)$. Our result generalizes their results to the class of all overlap graphs which includes k -nearest neighbor graphs.

We obtain a Mader-like result for overlap graphs. A classical result of Mader [6] states that if a graph G does not have a minor isomorphic to K_h , then no minor of G has average degree $\Omega(h \log h)$. We show that for all L , any depth L minor of an overlap graph in d dimensions has average degree bounded from above by $O(L^d)$.

We also consider some probabilistic properties of certain classes of random geometric graphs. We show that, with high probability, the Delaunay diagram, the relative neighborhood graph, and the k -nearest neighbor graph of a random point set excludes K_h as a depth L minor if $h = \Omega(L^{d/2} \log n)$.

2. Overlap graphs and shallow minors

Definition 2.1 [17]. Let $P = \{p_1, \dots, p_n\}$ be points in $\mathbb{R}^d$. A k -ply neighborhood system for P is a set, $\{B_1, \dots, B_n\}$, of closed balls such that (1) B_i is centered at p_i and (2) no point $p \in \mathbb{R}^d$ is strictly interior to more than k balls from B .

We will use the following notation. For each positive real α , if B is a ball of radius r in $\mathbb{R}^d$, then $\alpha \cdot B$ denotes the ball with the same center as B but radius αr .

Let $\Gamma = \{B_1, \dots, B_n\}$ be a k -ply neighborhood system. The *intersection graph* of Γ is the undirected graph with vertices $V = \{1, \dots, n\}$ and edges $E = \{(B_i, B_j) : (B_i \cap B_j) \neq \emptyset\}$.

Let $\alpha \geq 1$. The α -overlap graph of Γ is the undirected graph with vertices $V = \{1, \dots, n\}$ and edges

$$E = \{(B_i, B_j) : (B_i \cap (\alpha \cdot B_j) \neq \emptyset) \text{ and } ((\alpha \cdot B_i) \cap B_j \neq \emptyset)\}.$$

Clearly, when $\alpha = 1$, an overlap graph becomes the intersection graph of the k -ply neighborhood system.

We now define the k -nearest neighbor graph [24]. Let $P = \{p_1, \dots, p_n\}$ be a set of points in $\mathbb{R}^d$. The *nearest neighbor* of p_i is a point p_j , $j \neq i$, with minimum Euclidean distance from p_i . To make the nearest neighbor unique, we choose the point p_j with maximum index in case of ties and denoted it by $\text{nn}(p_i)$. Similarly, for any integer $k > 0$, let $\text{nn}_k(p_i)$ be the set of k nearest neighbors of p_i in P (again, use indices to break ties). The k -nearest neighbor graph of P , denoted by $k\text{-NNG}(P)$, is the directed graph (P, E) where $E = \{(p_i, p_j) \mid p_j \in \text{nn}_k(p_i)\}$.

We remark that overlap graphs include $k\text{-NNGs}$, well-shaped finite element and finite difference meshes, and planar graphs [18].

We now give the definition of shallow minors as introduced in [23]. Suppose G is an undirected graph. Let $G_1, G_2, \dots, G_n$ be h disjoint connected subgraphs of G . The *minor* defined by $G_1, G_2, \dots, G_h$ is a graph H whose vertex set is $\{1, \dots, h\}$ and whose edge set contains exactly all those pairs (i, j) : $1 \leq i, j \leq h$ such that there is an edge in G that connects a vertex of G_i to a vertex of G_j . Clearly, H can be obtained from a subgraph of G by contraction. The *depth* of such an H -minor is the maximum diameter of $G_1, \dots, G_h$.

3. A combinatorial lemma of geometric graphs

In this section, we show that the intersection graph and the α -overlap graph of a k -ply neighborhood system exclude K_h as a depth L minor whenever $h > k(4L + 1)^d$ and $h > k(4\alpha L + 1)^d$, respectively.

3.1. Shallow minors of intersection graphs

First, we prove a geometric lemma about neighborhood systems.

Lemma 3.1. *Suppose $\{B_1, \dots, B_n\}$ is a k -ply neighborhood system in $\mathbb{R}^d$. For each d -dimensional ball B with radius r , for all constant $\beta > 0$,*

$$|\{i : B_i \cap B \neq \emptyset \text{ and } r_i \geq \beta r\}| \leq (1 + 2\beta)^d \beta^{-d} k,$$

where r_i is the radius of B_i .

Proof. Let p be the center of B . Let $B' = (1 + \beta) \cdot B$.

We distinguish two types of balls that intersect B . The first class contains those balls whose centers are in B' and the second class contains those whose centers are in the exterior of B' .

For the first class, we can shrink each ball to a concentric one of radius βr . The shrinking operation maintains the k -ply condition of the resulting neighborhood system. So, without loss of generality, we assume that all balls in the first class have radius βr .

Now, let B_i be a ball of the second class of radius r_i and center p_i . Let q_i be the common point of the boundary of B' and the line passing through pp_i . Let B'_i be a ball centered at q_i of radius βr . Clearly B'_i is completely contained in B_i because B_i intersects B . Replacing B_i by B'_i still satisfies

the k -ply condition. So, without loss of generality, we assume all balls of the second class have the properties that their centers are on the boundary of B' and their radii are equal to βr .

Therefore, all balls that intersect B are completely contained in $(1 + 2\beta) \cdot B$. Suppose there are m of them, by the volume bound we have

$$mv_d(\beta r)^d \leq kv_d(1 + 2\beta)^d r^d,$$

where v_d is the volume of a unit d -dimensional ball, implying $m \leq k(1 + 2\beta)^d / \beta^d$. $\square$

Theorem 3.2. *Suppose Γ is a k -ply neighborhood system and G is the intersection graph of Γ . Then for all L , G excludes K_h as a depth L minor for $h > k(4L + 1)^d$.*

Proof. Suppose G has a K_h minor of depth L . Then there are h sets of balls, $\Gamma_1, \dots, \Gamma_h \subset \Gamma$ such that

- (1) For each i , the intersection graph of Γ_i is connected and its diameter is at most L .
- (2) For each pair i, j , there is a ball in Γ_i that intersects a ball in Γ_j .

Now, let B_i be the ball in Γ_i of the largest radius. Without loss of generality, let B_1 be the ball of the smallest radius among $\{B_1, \dots, B_h\}$. Assume that the radius of B_1 is r .

Because the diameter of the intersection graph of Γ_i is at most L and B_1 is the ball of the largest radius of Γ_1 , all balls of Γ_1 are completely contained in the ball $B' = (2L + 1) \cdot B_1$.

Therefore, for all $i > 1$, there is a ball from Γ_i that intersects B' .

We claim that for each $i > 1$, there is a ball in Γ_i of radius at least r that intersects $B'' = (4L - 1) \cdot B_1$. Here is the argument. Note first that the diameter of the intersection graph of Γ_i is at most L and there is a ball from Γ_i that intersects B' . If that ball has radius at least r , then we are done with Γ_i . If not, as we increase the radius of B' by $2r$, we will include the ball completely and meet balls from the next level. We repeat the argument. We terminate in less than $L + 1$ iterations, since we must meet either B_i (which has radius of at least r) or some other balls in Γ_i that have radius at least r .

Notice that the radius of B'' is $(4L - 1)r$. With $\beta = 1/(4L - 1)$, using Lemma 3.1, we have $h \leq (1 + 2\beta)^d \beta^{-d} k = (4L + 1)^d k$. $\square$

3.2. Shallow minors of overlap graphs

Theorem 3.3. *Suppose Γ is a k -ply neighborhood system and G is the α -overlap graph of Γ . Then for all L , G excludes K_h as a depth L minor for $h = k(4\alpha^d L + 1)^d$.*

Proof. The proof is similar to that of Theorem 3.2. Suppose G has a K_h minor with depth L . Then there are h sets of balls, $\Gamma_1, \dots, \Gamma_h \subset \Gamma$ such that (1) for each i , the α -overlap graph of Γ_i is connected and its diameter is at most L , and (2) for each pair i, j , there exist U_i in Γ_i and U_j in Γ_j such that $U_i \cap (\alpha \cdot U_j) \neq \emptyset$ and $(\alpha \cdot U_i) \cap U_j \neq \emptyset$.

Now, let B_i be the ball in Γ_i of the largest radius. Without loss of generality, let B_1 be the ball of the smallest radius among $\{B_1, \dots, B_h\}$. Assume that the radius of B_1 is r .

Because the diameter of the α -overlap graph of Γ_i is at most L and B_1 is the ball of the largest radius of Γ_1 , all balls of Γ_1 are completely contained in the ball $B' = (2\alpha L + 1) \cdot B_1$.

Therefore, for all $i > 1$, there is a ball U_j from Γ_i and a $U \in \Gamma_1$ such that $\alpha \cdot U$ intersects U_j .

We claim that for each $i > 1$, there is a ball in Γ_i of radius at least r that intersect $B'' = (4\alpha L - 1) \cdot B_1$. The argument is very similar to that for the intersection graph.

Notice that the radius of B'' is $(4\alpha L - 1)r$. With $\beta = 1/(4\alpha L - 1)$, using Lemma 3.1, we have $h \leq (1 + 2\beta)^d \beta^{-d} k = (4\alpha L)^d k$. $\square$

4. Random geometric graphs

In this section, we show that if P is a “random” point set of n points in d dimensions, then with probability approaching to 1, various types of geometric graphs, such as the Delaunay diagram, the k -nearest neighbor graph ($k \leq \Theta(\log n)$), the α -overlap graph, and the relative neighborhood graph of P exclude K_h as a depth L minor when $h = \Omega((L)^{d/2} \log n)$. Such result can also be generalized to random point sets on the surface of unit d -sphere in $\mathbb{R}^{d+1}$.

Formally, we use the homogeneous Poisson point process of intensity one as our probabilistic model. This standard model is characterized by the property that the number of points in a region is a random variable that depends only on the d -dimensional volume of the region [4,13,16,28]. In this model,

- The probability of exactly k points appearing in any region of volume V is $e^{-V} V^k / k!$.
- The conditional distribution of points in any region given that exactly k points fall in the region is uniform.

We denote the probability of an event by $\text{Prob}[\cdot]$.

4.1. Shallow minors for uniform neighborhood systems

Let $\beta \geq 1$ a constant. A k -ply $\{B_1, \dots, B_n\}$ is β -uniform if for all $i, j, r_i \leq \beta r_j$, where r_i is a radius of B_i for all $1 \leq i \leq n$.

Lemma 4.1. *Suppose Γ is a β -uniform k -ply neighborhood system and G is the intersection graph of Γ . Then for all L , G excludes K_h as a depth L minor for $h > k(4\beta L + 1)^{d/2}$.*

Proof. Let $\Gamma_1, \dots, \Gamma_h \subset \Gamma$ be defined as in the proof of Theorem 3.2.

We first note that there are at least $\Theta(h^2/k)$ balls in the union of Γ_i 's. This follows from the observation that the intersection graph of a k -neighborhood system of size n has at most $O(kn)$ edges [27] and there are at least h^2 edges in the intersection graph of $\Gamma_1 \cup \dots \cup \Gamma_h$ because they define a K_h -minor.

Then the argument is similar to that in the proof of Theorem 3.2. Let B be the smallest ball in $\Gamma_1 \cup \dots \cup \Gamma_h$, we can show that $(4\beta L + 1) \cdot B$ contains $\Gamma_1 \cup \dots \cup \Gamma_h$. Therefore, $h^2/k \leq k(4\beta L + 1)^d$, giving $h \leq O(kL^{d/2})$. $\square$

Theorem 4.2. *With high probability, for all L , a random k -NNG excludes K_h as a depth L minor for $h = \Omega(L^{d/2} \max(k, \log n))$.*

The basic idea is to show that with high probability there is a constant β that the k -NNG of a random point set is a subgraph of the intersection graph of a β -uniform neighborhood system. We use the following lemma of [28] which describes some probabilistic properties of random k -NNGs in $\mathbb{R}^d$.

Lemma 4.3. *There are constants $c, c_1 < c_2$ depending only on d such that with high probability (e.g., $1 - 1/n$), the radius of all balls in a $(c \log n)$ -nearest neighbor graph of a random point set P is between $c_1(\log^{1/d} n)$ and $c_2(\log^{1/d} n)$.*

Proof. Teng and Yao [28] gave a proof of this lemma for $d = 2$. Their proof can be directly extended to higher dimension. For completeness, we present here such a high-dimension extension.

Let $k = c \log n$ for a constant c to be specified. Let v_d be the volume of the unit d -dimensional ball.

Let $r = (\lambda k / v_d)^{1/d}$. Denote by $B(r)$ a ball with radius r centered at a point p of P , and by $|B(r)|$ the number of points of P contained in $B(r)$. Let r_k be the radius of k -nearest neighbor ball of p .

$$\text{Prob}[r_k > r] = \text{Prob}[|B(r)| < k] = e^{-\lambda k} \sum_{j < k} \frac{(\lambda k)^j}{j!}.$$

If $\lambda > 1$, then the terms in the summation strictly increase with j , hence

$$\text{Prob}[r_k > r] < k e^{-\lambda k} \frac{(\lambda k)^k}{k!} \leq k (\lambda e)^k e^{-\lambda k} = k (\lambda e^{-(\lambda-1)})^k.$$

Therefore, there is a constant c_2 such that $\text{Prob}[r_k > c_2 \log^{1/d} n] \leq 1/n^2$. So with probability at least $1 - 1/n$, the radius of all k -nearest neighbor ball is at most $c_2 \log^{1/d} n$.

For a lower bound on r_k , we have

$$\text{Prob}[r_k < r] = \text{Prob}[|B(r)| > k] = e^{-\lambda k} \sum_{j > k} \frac{(\lambda k)^j}{j!}.$$

For $\lambda < 1$, the terms in the summation decrease faster than a geometric series with ratio λ , hence

$$\text{Prob}[r_k < r] \leq \frac{1}{1-\lambda} e^{-\lambda k} \frac{(\lambda k)^k}{k!} \leq \frac{1}{1-\lambda} (\lambda e^{1-\lambda})^k.$$

Therefore, there is a constant c_1 such that $\text{Prob}[r_k < c_1 \log^{1/d} n] \leq 1/n^2$. So with probability at least $1 - 1/n$, the radius of all k -nearest neighbor ball is at least $c_1 \log^{1/d} n$. $\square$

In [28], Teng and Yao showed that with high probability, all edges in the Delaunay diagram of a random points are “short”, and hence contained in the $\Theta(\log n)$ -NNG. The following corollary is a direct consequence of Lemma 4.3.

Corollary 4.4. *There exists a constant c depending only on d such that with high probability (e.g., $1 - 1/n$), Delaunay diagram is a subgraph of the $(c \log n)$ -nearest neighbor graph of a set P of random points.*

We now prove Theorem 4.2. We first note that the property of excluding shallow minor is closed under the subgraph operation.

Proof of Theorem 4.2. By Lemma 4.3, there exist a pair of constants c_1 and c_2 such that with high probability, the radii of all k -nearest neighbor balls are between $c_1(\log^{1/d} n)$ and $c_2(\log^{1/d} n)$. This implies that with high probability, the resulting neighborhood system is β -uniform for $\beta = c_2/c_1$. The theorem then follows from Lemma 4.1. $\square$

Corollary 4.5. *With very high probability, for all L , the Delaunay diagram and the relative neighborhood graph of a random point exclude K_h as a depth L minor whenever $h = \Omega(L^{d/2} \log n)$.*

Proof. It follows from Corollary 4.4 and Theorem 4.2. $\square$

5. Average degree of shallow minors in geometric graphs

A classical result of Mader [6] states: There is a constant c such that if a graph G does not have a minor isomorphic to K_h , then no minor of G has the average degree more than $ch \log h$ for a constant c .

What would be a Mader-like result for shallow minors? The first guess may be: If a graph G does not have a depth L minor isomorphic to the h -clique, then all depth L minors of G have average degree bounded from above by a polynomial in h and L . Unfortunately, this is not true. For example,² take a complete n by n bipartite graph; clearly the average degree of this graph is n . But it does not have a depth 0 minor isomorphic to K_3 .

We now show that a “stronger version” of Mader-like result does exist for our geometric graphs.

Theorem 5.1. *Suppose G is the intersection graph of a k -ply neighborhood system Γ in $\mathbb{R}^d$ and H is a depth L minor of G . Then the average degree of H is bounded from above by $O(kL^d)$.*

Proof. Suppose H has m vertices $\{h_1, \dots, h_m\}$. Because H is a depth L minor of G , there are m disjoint subsets $\Gamma_1, \dots, \Gamma_m$ of Γ that define H where Γ_i corresponds to h_i . Let B_i be the ball of the largest radius of Γ_i . Let r_i be the radius of B_i . Without loss of generality, we assume that $r_1 \leq r_2 \leq \dots \leq r_m$. Direct each edge of H from its lower indexed vertex to its higher indexed vertex. We claim that the outdegree of the resulting directed graph is bounded by $O(kL^d)$. The theorem then follows. The proof to the claim above is similar to that of Theorem 3.2. $\square$

Similarly, we have Theorem 5.2.

Theorem 5.2. *Suppose G is the α -overlap graph of a k -ply neighborhood system in $\mathbb{R}^d$ and H is a depth L minor of G . Then the average degree of H is bounded from above by $O(k(\alpha L)^d)$.*

Using a similar argument, we can also show that, with very high probability, for all L , the average degree of any depth L minor of a random k -NNG is bounded above by $O(L^{d/2} \max(k, \log n))$. Moreover, the Delaunay diagram and the relative neighborhood graph of a random set of points in $\mathbb{R}^d$ have no depth L minor of average degree more than $\Omega(L^{d/2} \log n)$.

6. Applications

Our shallow minor results make it possible to apply some known combinatorial construction to overlap graphs. In particular, we can use the algorithm of Plotkin et al. [23] to decompose overlap

² Given by Yuan Ma of Department of Mathematics, MIT.

graphs and k -NNGs for a linear or almost linear speed-up emulation of overlap graphs on hypercubes and butterflies, respectively. Such emulations can be used to design efficient parallel finite element and finite difference algorithms and parallel computational geometry algorithms for k -NNG related problems.

6.1. Cut-cover for overlap graphs and k -NNG

Suppose $G = (V, E, w)$ is a node weighted undirected graph where w is the weight function. For each vertex $u \in V$ let $N^t(u)$ be the set of all vertices in V that are within (combinatorial) distance t from u . A t -neighborhood cover of G , as defined by Awerbuch and Peleg [1] is a set of (possible overlapping) subsets $\{S_1, \dots, S_k\}$ of G such that for all $u \in V$, there exists an i , such that $N^t(u) \subseteq S_i$.

Suppose $C \subseteq V$ is a subset of V , by G/C we denote the graph $(V - C, E \cap (V - C) \times (V - C))$.

The *balanced cut-cover*, defined by Kaklamanis et al. [12], combines the notion of t -neighborhood covers and small separators. Formally, for $0 < \beta < 1$, $s, t, \varepsilon \in \mathbb{R}^+$, a β -balanced (t, s, ε) cut-cover of G is a set $C \subseteq V$ and a t -neighborhood cover $\{S_1, \dots, S_l\}$ of G/C such that

- (1) $|C| \leq s$;
- (2) $\sum_{i=1}^l w(S_i) \leq (1 + \varepsilon)w(V)$; and
- (3) $\max_{i=1}^l w(S_i) \leq \beta w(V)$.

A q -color β -balanced (t, s, ε) cut-cover of an n node graph $G = (V, E)$ where nodes are colored with q colors is a set C and a t -neighborhood cover $\{S_1, \dots, S_l\}$ of G/C that is a β -balanced (t, s, ε) cut-cover of G for every weight function defined by an indicator function for the q colors of G , ignoring other colors.

Kaklamanis, Krizanc and Rao showed that planar graphs and graphs with bounded forbidden minors have good cut-covers and posed an open question of whether intersection graphs and overlap graphs have “good” cut-covers.

The following theorem follows from Theorem 3.3 and the construction of Plotkin et al. [23]. The time complexity for finding such a cut-cover is $O(k\alpha^d n^2/L)$.

Theorem 6.1. Suppose $G = (V, E)$ is an α -overlap graph of a k -neighborhood system in $\mathbb{R}^d$. Then for all integers q , t and $L > 2qt$, and q coloring functions on G , there is a set $C \subset V$ and a t -neighborhood cover $\{S_1, \dots, S_l\}$ such that

$$|C| = O(q^2 \alpha^{d-1} k L^{2d-1} \log n),$$

$$\sum_{i=1}^l w(S_i) \leq (1 + 2qt/L)w(V),$$

$$\max_{i=1}^l w(S_i) \leq (2/3)w(V),$$

with respect to each of the q colors function w ignoring the other $q - 1$ colors.

6.2. Fast emulations in hypercubes and butterflies

Our result that overlap graphs have “good” cut-cover yields an immediately application to parallel computation on hypercubic and butterfly networks, thanks to the result of Kaklamanis et al. [12].

Theorem 6.2. *Bounded degree α -overlap graphs can be emulated on hypercube graphs with a constant factor slow down and on butterfly graphs with a factor of $O(\log^* n)$ slow down.*

The emulation result can be applied to parallel finite element and finite difference computations and computational geometry problems that use k -NNGs.

For example, a natural condition for well-shaped meshes for the adaptive finite difference method is the density of the embedding [2,21]. Let G be an undirected graph and let π be an embedding of its nodes in $\mathbb{R}^d$. We say π is an embedding of *density* α if the following inequality holds for all vertices v in G . Let u be the closest node to v . Let w be the farthest node from v that is connected to v by an edge. Then

$$\frac{\|\pi(w) - \pi(v)\|}{\|\pi(u) - \pi(v)\|} \leq \alpha.$$

In general, G is an α -density graph in $\mathbb{R}^d$ if there exists an embedding of G in $\mathbb{R}^d$ with density α . It can be easily shown that there is a $\Delta(\alpha, d)$ depending only on α and d such that the maximum degree of an α -density graph is bounded by $\Delta(\alpha, d)$.

Corollary 6.3. *Density graphs and k -NNGs can be emulated on hypercube graphs with a factor of constant slow down and on butterfly graphs with a factor of $O(\log^* n)$ slow down. Therefore, iterative methods for solving a finite difference system of a PDE (on a well-conditioned adaptive mesh) can be performed on a hypercubic parallel machine with a linear speed-up.*

Recently, Leiserson et al. [14] showed that “good” cut-cover implies better out-of-core algorithms for iterative relaxation algorithms on the graphs. Our results can be used to extend their tools for solving adaptive finite difference systems.

7. Final remarks and open questions

In this paper, we show that certain classes of geometric graphs have some common combinatorial properties. Our results, to a certain degree, complement the work of Miller et al. [17,18] which can be viewed as an attempt to find a geometric characterization for a combinatorial property, namely, the existence of small separators. This paper tries, in some sense, to understand the combinatorial aspects of geometric graphs. The goal of such an understanding is for

- applying combinatorial construction to geometric graphs, and
 - designing efficient combinatorial algorithms using our knowledge from (computational) geometry.
- An open question is whether the $\Omega(L^d)$ bound can be reduced to $\Omega(L^{d/2})$. We notice that by amortizing the steps and volume, we are able to reduce the exponent of L from d to $d - 1$. We conclude the paper by giving a simple argument to reduce the exponent of L from d to $d - 0.5$.

In the proof of Theorem 3.2, we showed that for each $i > 1$, there is a ball B'_i in Γ_i of radius at least r that intersects B'' . We divide B'_i into two classes: C_1 contains balls which are less than $D = L^{(d-0.5)/d}r$ away from the center of B'' ; C_2 contains all other balls. By volume argument, there are at most $O(L^{d-0.5})$ number of balls in C_1 . We now show that the number of balls in C_2 is bounded by $O(L^{d-0.5})$. The distance between B'_i and the center of B'' is at least D , and at most $2L$ balls connecting B_1 to B'_i . The total volume of these ball is at least $L^{0.5}$. Therefore, by volume argument, there are at most $O(L^d/L^{0.5})$ balls in C_2 . Fine tuning this argument further reduces the exponent to $d - 1$. However, we conjecture that the correct value of the exponent is $d/2$.

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